



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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CSE - Academic Year 2026-27
Java Programming Lab | Batch 2025-29

ASSIGNMENT : 20	
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Batch	CSE A - 2
Class	Java Programming Lab
Academic Year & Semester	2025-29 : Semester 3
Date of Submission	

TITLE : Develop a Java application to establish connectivity with a database and perform INSERT, UPDATE, and DELETE operations.

Learning Objectives :

To perform database modification operations using JDBC.

Theory :

JDBC (Java Database Connectivity) enables Java applications to interact with relational databases & perform database operations using SQL statements. INSERT, UPDATE and DELETE are DML operations used to modify data stored. The INSERT statement is used to add new records, UPDATE statement modifies existing records, and DELETE statement removes records from a table. These can be performed using the **Statement** or **PreparedStatement** interface.

The executeUpdate() method is used to execute these operations and returns the number of records affected. JDBC provides a simple and efficient way to manage

database records and is widely used in applications such as student management, banking, and inventory systems.

Exercise :

1. Create a Java application to manage **employee records** using CRUD operations.

```

import java.sql.*; public class java20a {      public
static void main(String[] args) {          try {
Connection con = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/jdbc_lab",
        "root",
        "anurag" );

        Statement st = con.createStatement();
st.executeUpdate(        "insert into employee values (101,
'Anushree', 'CSE',
50000),(102, 'Natasha', 'IT', 20000)"
        );

        ResultSet rs = st.executeQuery("select * from employee");

        System.out.println("Records inserted :");
        System.out.println("Employee Records");
System.out.println("-----");
while (rs.next()) {          System.out.println(
rs.getInt("emp_id") + "  " +
rs.getString("name") + "  " +
rs.getString("department") + "  " +
rs.getInt("salary")          );
        }
        System.out.println();          st.executeUpdate(
"update employee set salary = 55000 where emp_id = 102"
        );

```

```

        rs = st.executeQuery("select * from
employee");
        System.out.println("Record updated :");
        System.out.println("Employee Records");
        System.out.println("-----");
        while (rs.next()) {
            System.out.println(
rs.getInt("emp_id") + " " +
rs.getString("name") + " " +
rs.getString("department") + " " +
rs.getInt("salary")
            );
        }
        System.out.println();
        st.executeUpdate(
            "delete from employee
where emp_id = 101"
        );
        rs = st.executeQuery("select *
from employee");
        System.out.println("Record deleted :");
        System.out.println("Employee Records");
        System.out.println("-----");
        while (rs.next()) {
            System.out.println(
rs.getInt("emp_id") + " " +
rs.getString("name") + " " +
rs.getString("department") + " " +
rs.getInt("salary")
            );
        }
        con.close();
    } catch
(Exception e) {
        System.out.println(e);
    }
}
}

```

In SQL :

```
• use jdbc_lab;
• create table employee (
    emp_id int,
    name varchar(50),
    department varchar(50),
    salary double
);
```

OUTPUT :

Records inserted :

Employee Records

101 Anurag CSE 50000

102 Natasha IT 20000

Record updated :

Employee Records

101 Anurag CSE 50000

102 Natasha IT 55000

Record deleted :

Employee Records

102 Natasha IT 55000

PS C:\Users\softlink\Desktop\java code>

2. Create a Java application to manage **student records** using **CRUD operations**.
Perform Create, Read, Update, and Delete operations on student details such as roll number, name, course, and marks using a database.

```

import java.sql.*; public class java20b {
public static void main(String[] args) {
    try {
        Connection con =
DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/jdbc_lab",
            "root",
            "anurag" );

        Statement st = con.createStatement();
        // INSERT
st.executeUpdate(
            "insert into
student values " +
            "(101, 'Anurag', 'CSE', 93)," +
            "(102, 'Natasha', 'IT', 85)" );

        System.out.println("Records inserted");
        System.out.println("Student Records");
        System.out.println("-----");
        ResultSet rs = st.executeQuery("select * from
student");
        while (rs.next()) {
System.out.println(
            rs.getInt("roll_no") + " "
+
            rs.getString("name") + " " +
rs.getString("course") + " " +
rs.getInt("marks")
        );
    }

    // UPDATE
    st.executeUpdate(
"update student set marks = 90 where roll_no = 102" );

```

```

        System.out.println("\nRecord updated");
        System.out.println("Student Records");
        System.out.println("-----");
rs = st.executeQuery("select * from student");           while
(rs.next()) {           System.out.println(
rs.getInt("roll_no") + " " +
rs.getString("name") + " " +
rs.getString("course") + " " +
rs.getInt("marks")           );
        }
        // DELETE           st.executeUpdate(
"delete from student where roll_no = 101" );

        System.out.println("\nRecord deleted");
        System.out.println("Student Records");
        System.out.println("-----");
rs = st.executeQuery("select * from student");           while
(rs.next()) {           System.out.println(
rs.getInt("roll_no") + " " +
rs.getString("name") + " " +
rs.getString("course") + " " +
rs.getInt("marks")           );           }
con.close();           } catch (Exception e) {
        System.out.println(e);
    }
}
}

```

In SQL :

```
• use jdbc_lab;
• create table student (
    roll_no int,
    name varchar(50),
    course varchar(50),
    marks int
);
```

OUTPUT :

```
SOFTLINK\AppData\Local\Temp\cp_4to120505958
Records inserted
Student Records
-----
101  Anurag   CSE   93
102  Natasha  IT    85

Record updated
Student Records
-----
101  Anurag   CSE   93
102  Natasha  IT    90

Record deleted
Student Records
-----
102  Natasha  IT    90
PS C:\Users\softlink\Desktop\java code>
```

Outcome :

- Understood database modification operations using JDBC.
- Learned to insert, update, and delete records using SQL statements.
- Learned to retrieve records using the SELECT operation.
- Understood the use of executeUpdate() and ResultSet in JDBC.
- Successfully performed basic CRUD operations on database records.

Conclusion :

This assignment demonstrated database modification and retrieval operations using JDBC. Employee and student records were successfully inserted, retrieved, updated, and deleted using SQL statements. The assignment helped in understanding the basic implementation of CRUD operations in a Java application connected to a database.